

Factorial Designs

Part 1

PUBH 526

Design and Analysis of Randomized Trials in Public Health

Nilupa S. Gunaratna, MS, PhD

Design of Experiments: Big Idea

- Two primary types of structure in an experiment
 - Treatment structure - factors of interest in study
 - Design structure, e.g., blocks, strata, clusters
- So far, we've focused on a one-way treatment structure under different design structures
 - CRD – parallel arms, no design structure
 - RCBD – one blocking factor
 - Latin square – two blocking factors
 - Cluster randomized trial - clustering
- Factorial design – two (or more) treatments

Linear Model Construction

- Construct model for y based on structures of experiment:

$$y = \text{design structure components} + \text{treatment structure components} + \text{error structure}$$

- Elements of **treatment structure** and **design structure** may be fixed or random
 - Treatment structure: more likely to be fixed
 - Design structure: more likely to be random
- The **error structure** is often derived from the interactions of the treatment structure with the design structure
- We'll now address more complex **treatment structures**

Two-factor CRD

- In one-factor CRD, treatments are often levels of one factor
- However, we may be interested in combinations of levels from two (or more) factors
 - Intervention on parents & intervention on kids for child milk consumption
 - Vitamin A & zinc for placental malaria
 - Diet & exercise interventions
- Could treat each combination as a treatment
 - a levels of factor A and b levels of factor B
 - ab total treatments
 - Use contrasts to study specific factor effects

Example

An experiment is conducted to study the effect of hormones injected into test rats. There are two distinct hormones (A,B) each with two distinct levels. We will consider this to be four different treatments labeled {A,a,B,b}. Each treatment is applied to six rats with the response being the amount of glycogen (in mg) in the liver.

- 4 parallel arms, 6 rats each
- “rat 2x2.sas”
- Contrasts for:
 - Hormone A vs. B?
 - High dose vs low dose?
 - Hormone effect the same at each dosage?
 - Effect of higher dose for hormone A vs. effect of higher dose for hormone B?

Example

- However, we're not dealing 4 distinct interventions
- We have specific hypotheses, as evidenced by our contrasts
- Can we analyze without using contrasts?

Factorial Experiment

- Consider treatments as level combinations of two factors
- In this example, each factor has 2 levels
- Treatment structure known as a 2 x 2 factorial (or 2^2 factorial)
 - Factor A: Hormone (A or B)
 - Factor B: Dose (Low or High)
- In general, a full factorial structure investigates all level combinations of the two factors (vs. a fractional factorial structure)
- If a levels of Factor A and b levels of Factor B, a single replicate of a factorial design involves ab observations

Statistical Model

- Statistical model (for CRD design structure) is

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \epsilon_{ijk} \quad \left\{ \begin{array}{l} i = 1, 2, \dots, a \\ j = 1, 2, \dots, b \\ k = 1, 2, \dots, n \end{array} \right.$$

can relax
this to be n_{ij}

τ_i - i th level effect of factor A (main effect - ignores B)

β_j - j th level effect of factor B (main effect - ignores A)

$(\tau\beta)_{ij}$ - interaction effect of combination ij

$(\tau\beta)_{ij}$ explains deviation of means from additive model

$\epsilon_{ijk} \sim N(0, \sigma^2)$

Model Estimates

- Under the “sum to zero” parameter constraints:

$$\hat{\mu} = \bar{y}_{...}$$

$$\hat{\tau}_i = \bar{y}_{i..} - \bar{y}_{...}$$

$$\hat{\beta}_j = \bar{y}_{.j.} - \bar{y}_{...}$$

$$(\widehat{\tau\beta})_{ij} = \bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...}$$

- The predicted value is ij combination average, so

$$\hat{y}_{ijk} = \bar{y}_{ij.} \text{ and } \hat{\epsilon}_{ijk} = y_{ijk} - \bar{y}_{ij.}$$

- We would get the same predicted values and residuals if we combined factors into single treatment factor and used one-way ANOVA

Note

- Based on linear model...

$$\begin{aligned}\bar{y}_{i..} &= \mu + \tau_i + \bar{\beta}_{.} + \overline{(\tau\beta)}_{i.} \\ \bar{y}_{i..} - \bar{y}_{i'..} &= \tau_i - \tau_{i'} + \overline{(\tau\beta)}_{i.} - \overline{(\tau\beta)}_{i'..}\end{aligned}$$

- Difference in Factor A levels depends on $(\tau\beta)$ constraints
- Thus, $\tau_i - \tau_{i'}$ is non-estimable (if interaction present)
- Caution testing main effects if interaction present
- Should always test interaction first and proceed to main effects only when appropriate!!!

Partitioning the Sum of Squares

- Rewrite each observation as

$$y_{ijk} = \bar{y}_{...} + (\bar{y}_{i..} - \bar{y}_{...}) + (\bar{y}_{.j.} - \bar{y}_{...}) + (\bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...}) + (y_{ijk} - \bar{y}_{ij.})$$

- The total SS $\sum (y_{ijk} - \bar{y}_{...})^2$ can be broken up into

$$\begin{aligned} & bn \sum_i (\bar{y}_{i..} - \bar{y}_{...})^2 + \\ & an \sum_j (\bar{y}_{.j.} - \bar{y}_{...})^2 + \\ & n \sum_i \sum_j (\bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...})^2 + \dots \end{aligned}$$

- $SS_A + SS_B + SS_{AB} + SS_E$
- Under normality, all SS/σ^2 independent

ANOVA Table

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
Factor A	SS_A	$a - 1$	MS_A	F_0
Factor B	SS_B	$b - 1$	MS_B	F_0
Interaction	SS_{AB}	$(a - 1)(b - 1)$	MS_{AB}	F_0
Error	SS_E	$ab(n - 1)$	MS_E	
Total	SS_T	$abn - 1$		

- Does this look suspiciously like the stratified randomization examples?

Comparison with RCBD?

- $df_E > 0$ only if $n > 1$
 - When $n = 1$, cannot separate interaction from error (confounded)
 - Recall RCBD sometimes uses $n = 1$. In those cases, assuming no interaction in RCBD allows us to estimate error and test for treatment differences.
- If $n > 1$, what's the difference between this and a RCBD?

Hypothesis Testing

- Can show: Fixed Case

$$E(MS_E) = \sigma^2$$

$$E(MS_A) = \sigma^2 + bn \sum \tau_i^2 / (a - 1)$$

$$E(MS_B) = \sigma^2 + an \sum \beta_j^2 / (b - 1)$$

$$E(MS_{AB}) = \sigma^2 + n \sum (\tau\beta)_{ij}^2 / (a - 1)(b - 1)$$

- Use F -tests to test equality of A, B, and AB effects

$$F_0 = \frac{SS_A / (a - 1)}{SS_E / (ab(n - 1))}$$

$$F_0 = \frac{SS_B / (b - 1)}{SS_E / (ab(n - 1))}$$

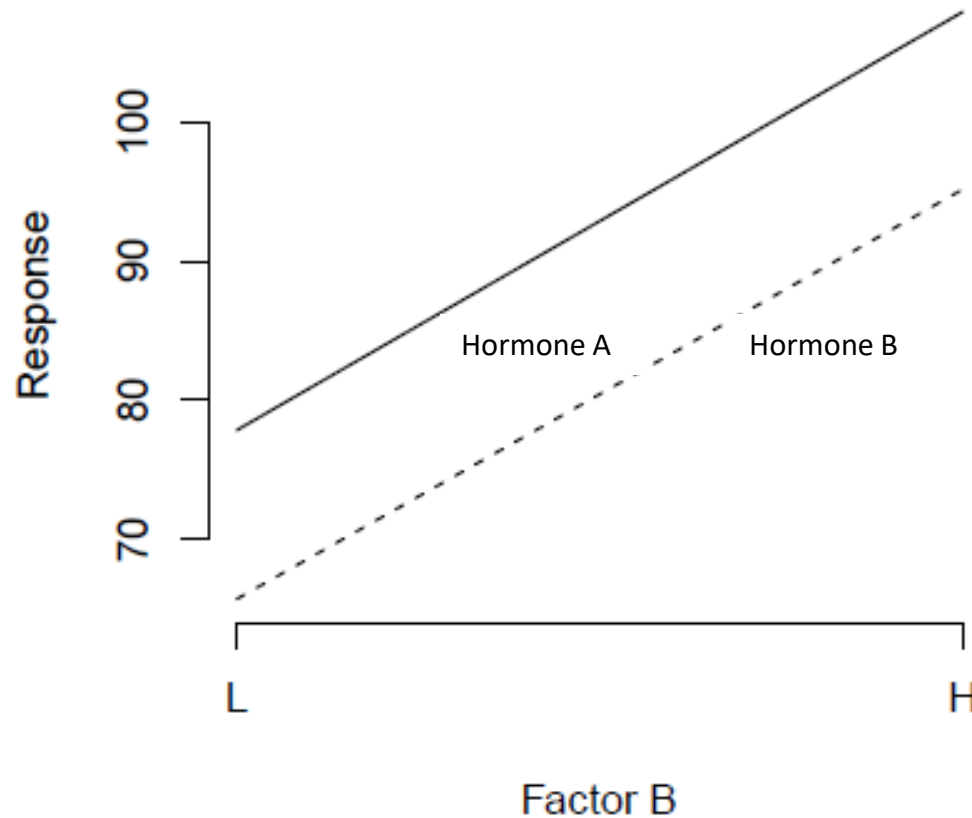
$$F_0 = \frac{SS_{AB} / (a - 1)(b - 1)}{SS_E / (ab(n - 1))}$$

Interpretation

- Interaction first
- If interaction not significant, the main effects for one factor average out the effects of the other factor
- Many studies will power on main effects (not necessarily interaction)

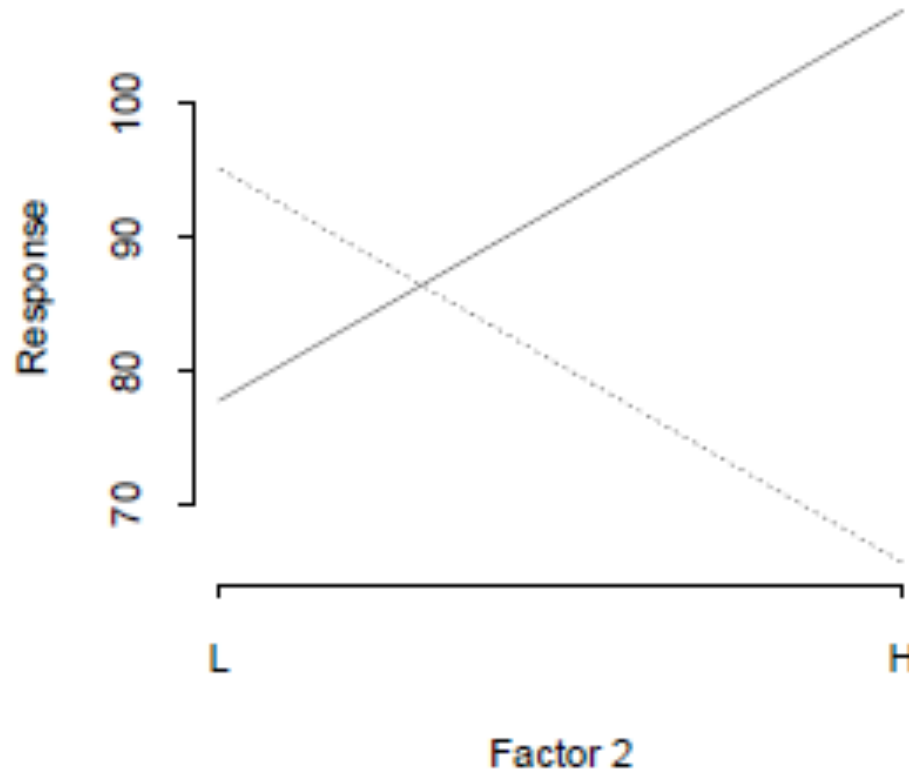
Interaction Plots

- No interaction



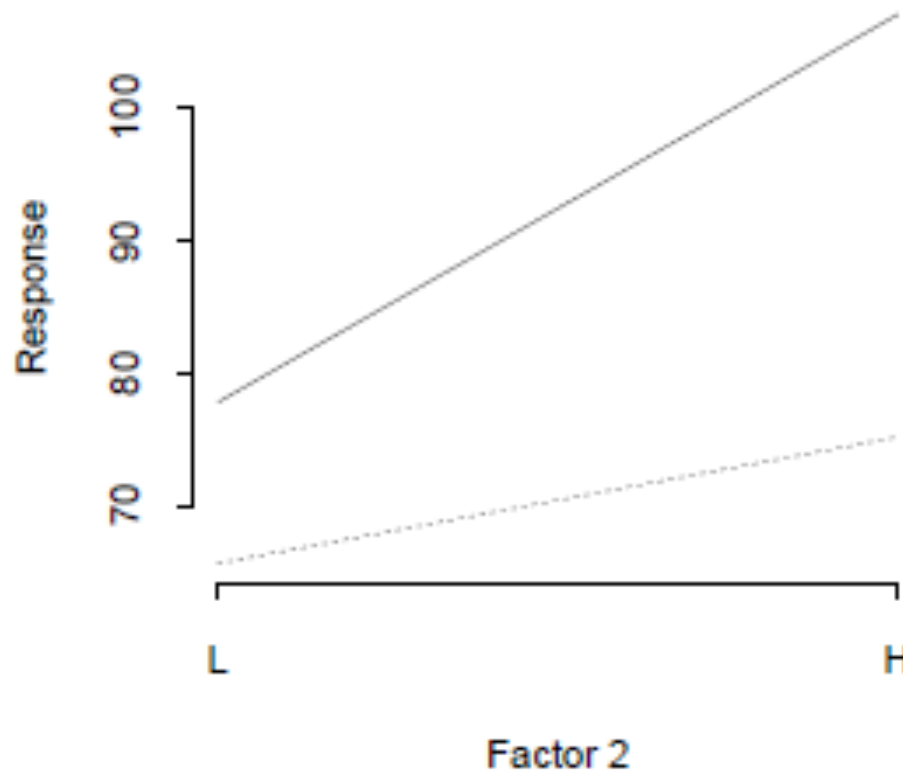
Interaction Plots

- Completely opposite effects



Interaction Plots

- Qualitatively, but not quantitatively, the same effect



Regression Approach

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_{12} x_{1i} x_{2i} + \epsilon$$

- How to code x_{1i} and x_{2i} ?
- What is the meaning of each β ?
 - Answer: it depends on how you code x_{1i} and x_{2i}
- What is the predicted value of y for each treatment combination?
- What happens if you have more than 2 levels of one or both factors?

Interaction Effects

- If $\beta_{12} \neq 0$, leave x_1 and x_2 in model
- In general, if interaction is significant, leave the main effects in the model
- “Main effect” then depends on the level of the other factor
 - Cannot simply average the other factor out (ignore it)
 - Example: opposite behavior $\rightarrow$ main effect “cancels” out
 - Common to compare y_{ij} means for each level of i or j
- Sometimes interaction may due to only a few treatment combos

Diagnostics

- General linear model assumptions
 - 1 Errors independent, normally distributed
 - 2 Constant variance
 - 3 No interaction (if $n = 1$)
- Normality

Histogram or QQplot of residuals
- Variance and Unusual Observations

Residuals vs Factor A

Residuals vs Factor B

Residuals vs $\hat{y}_{ij}$.

Multiple Comparisons

- Common to use significance level α for each of the three F-tests (main effects and interaction)
 - Argument based on these being pre-planned comparisons
- If interaction significant
 - Construct interaction plot for visual description
 - Compare levels of A at fixed level of B and vice versa
 - SAS means statement only does main effect comparisons (I tend to just avoid this)
 - SAS lsmeans statement has slice option (does group F test)
 - SAS lsmeans with Proc GLIMMIX has slicediff option (does t tests)